



Quantum Mirror Symmetry Via Gross-Siebert Program

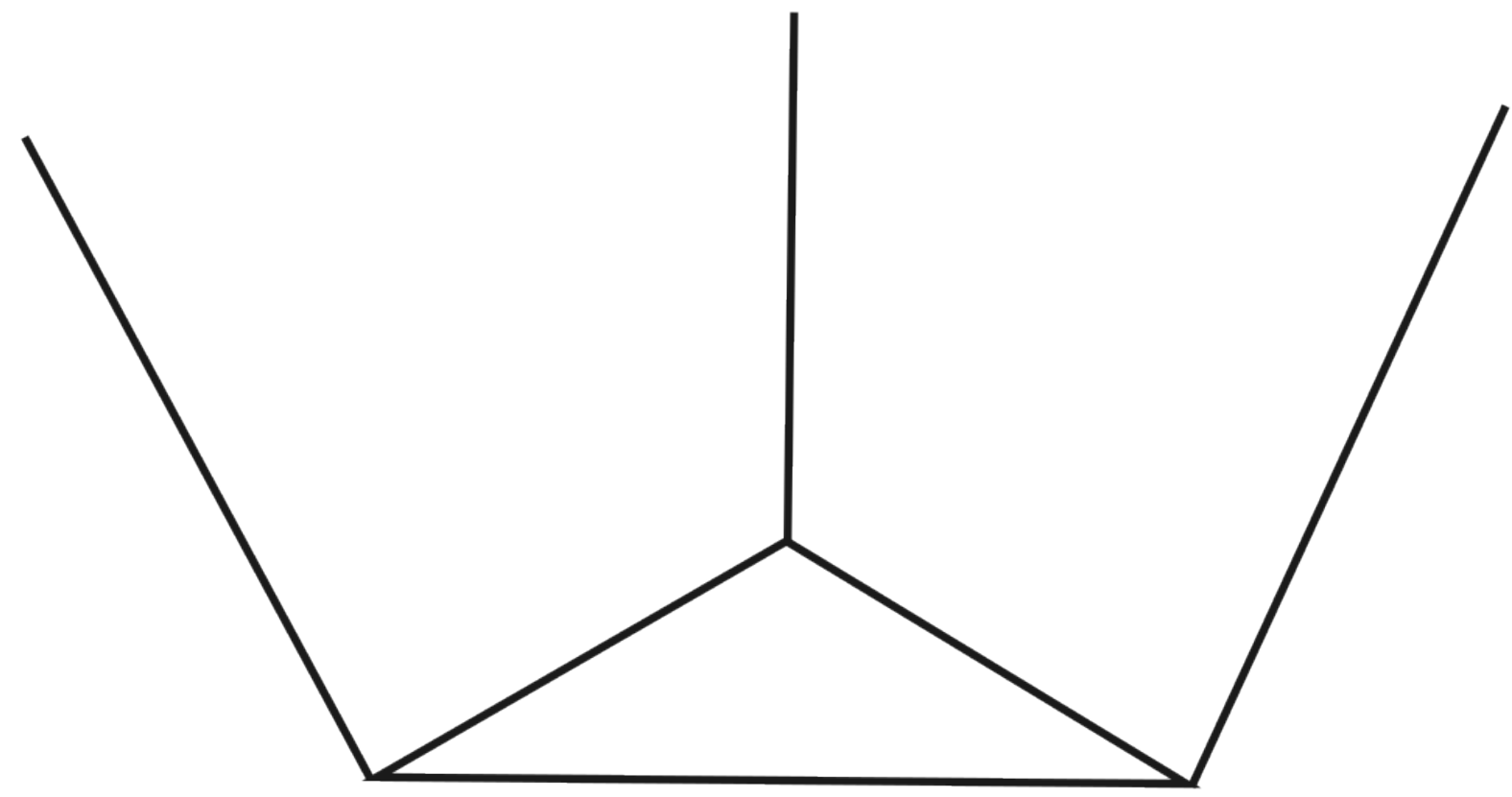


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Quantum A-Models on Toric Calabi-Yau Threefolds

e.g. $K_{\mathbb{P}^2}$: Total Space of Canonical Bundle of $\mathbb{P}^2$



- **Key Quantities of Quantum A-Models:** Higher Genus NS-limit of Closed Refined Topological String Theory A-Model of $K_{\mathbb{P}^2}$ = Higher Genus Logarithmic Gromov-Witten of $(\mathbb{P}^2, E)$ with λ_g -class Insertion [Bousseau].

Quantum B-Models on Hori-Vafa Mirror Landau-Ginzburg Models

$(T_{HV} := (\mathbb{C}^*)_{X,P}^2, W_{HV}), W_{HV} = T(X + P + \frac{1}{XP}),$
 W_{HV} Superpotential, T Complex Structure Modulus.

- Classical Mirror Curve $C := \{-1 + W_{HV} = 0\}$.
- Quantum Mirror Curve Equation
 $(-1 + \widehat{W}_{HV}(e^x, e^{\hbar\partial_x}))\psi(x) = 0$
 Given by Substitution $\widehat{X} := e^x, \widehat{P} := e^{\hbar\partial_x}$.
 e.g.
 $(-1 + Te^x)\psi(x) + T\psi(x + \hbar) + Te^{\frac{1}{2}\hbar}e^{-x}\psi(x - \hbar) = 0.$

- WKB Approximation: Ansatz

$$\psi(x) = \exp\left(\frac{1}{\hbar} \int^x \lambda_{\hbar}\right), \quad \lambda_{\hbar} = \sum_{i \geq 0} \lambda_i \hbar^i.$$

Plug into the Quantum Curve Equation to Get Recursive Relations on λ_i .

Quantum 1-form:

$$\lambda_{\hbar} = \log Pd \log X + \text{Quantum Correction } O(\hbar)$$

- **Key Quantities of Quantum B-Models:** Quantum Periods on the Mirror Curve C :

$$\oint \lambda_{\hbar}$$

Conjecture (Quantum Local Mirror Symmetry)

The Higher Genus NS-Limit of Closed Refined Topological String A-Model on a Toric Calabi-Yau Threefold is Equivalent to the Quantum Periods on the Mirror Curve in the Mirror Quantum Landau-Ginzburg Model. [Aganagic-Cheng-Dijkgraaf-Kreft-Vafa]

Main Theorem [Bousseau-Siebert-Y.]

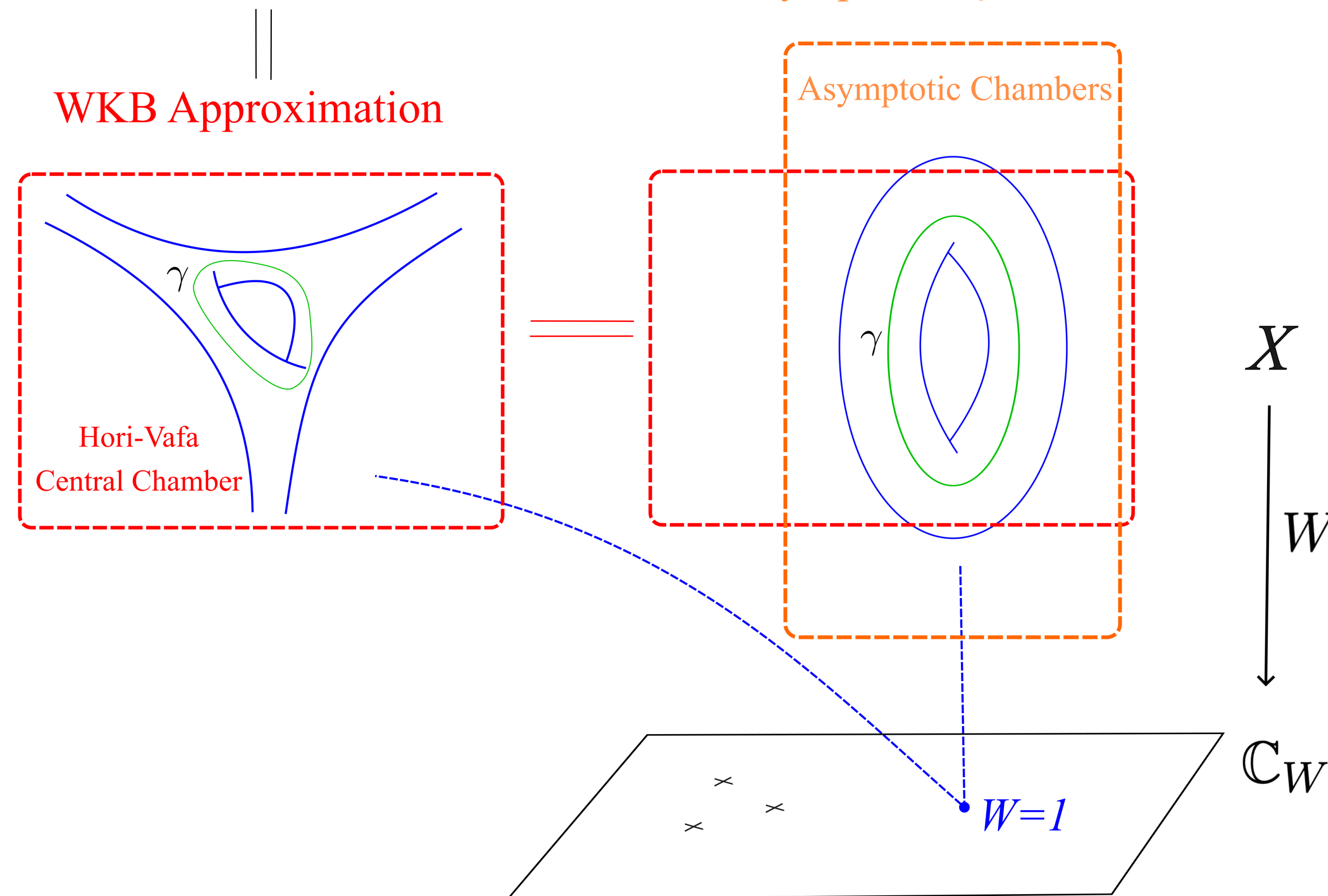
The Quantum Local Mirror Symmetry Holds in the Case of $K_{\mathbb{P}^2}$. (Our Method Could be Easily Generalized to the Canonical Bundles of All Toric del Pezzo Surfaces.)

Sketch of the Proof

Key Idea: To Compute the Intrinsically Defined Quantum Period $QC(\mathcal{M}_{LG})$ on Different Coordinates: on Both the Hori-Vafa Central Chamber and the Asymptotic Chambers in the Gross-Siebert Program.

Intrinsically Defined Quantum Periods
 $\langle QC(\mathcal{M}_{LG}), \gamma \rangle$

Quantum Hamiltonians of
 Asymptotic Quantum Walls



A Key Observation

- Different choices of toric coordinates $\widehat{X}, \widehat{P}$ will lead to different WKB approximations $\psi(x)$.
- The quantum periods, i.e. monodromies of $\psi(x)$, doesn't depend on the choice of coordinates!

Motivating Questions

- **Q:** Intrinsic (coordinate-free) interpretation of quantum periods? **A:** Deformation Quantization
- **Q:** How to prove quantum mirror symmetry after knowing the intrinsic interpretation of quantum periods? **A:** Gross-Siebert Program

Deformation Quantization [Kashiwara et al.]

Intrinsic NC Geometric Data on (T_{HV}, C) :

- Deformation Quantization of T_{HV} : A Sheaf of Noncommutative Algebras (Algebroid) $\mathcal{A}_{T_{HV}}$ on T_{HV} as a 1-parameter Deformation of $\mathcal{O}_{T_{HV}}$, e.g. Quantum Torus
- Deformation Quantization of C : A Left Simple Holonomic DQ Module Supported on C with Classical Limit $\mathcal{O}_C$, e.g. Landau-Ginzburg DQ-Module

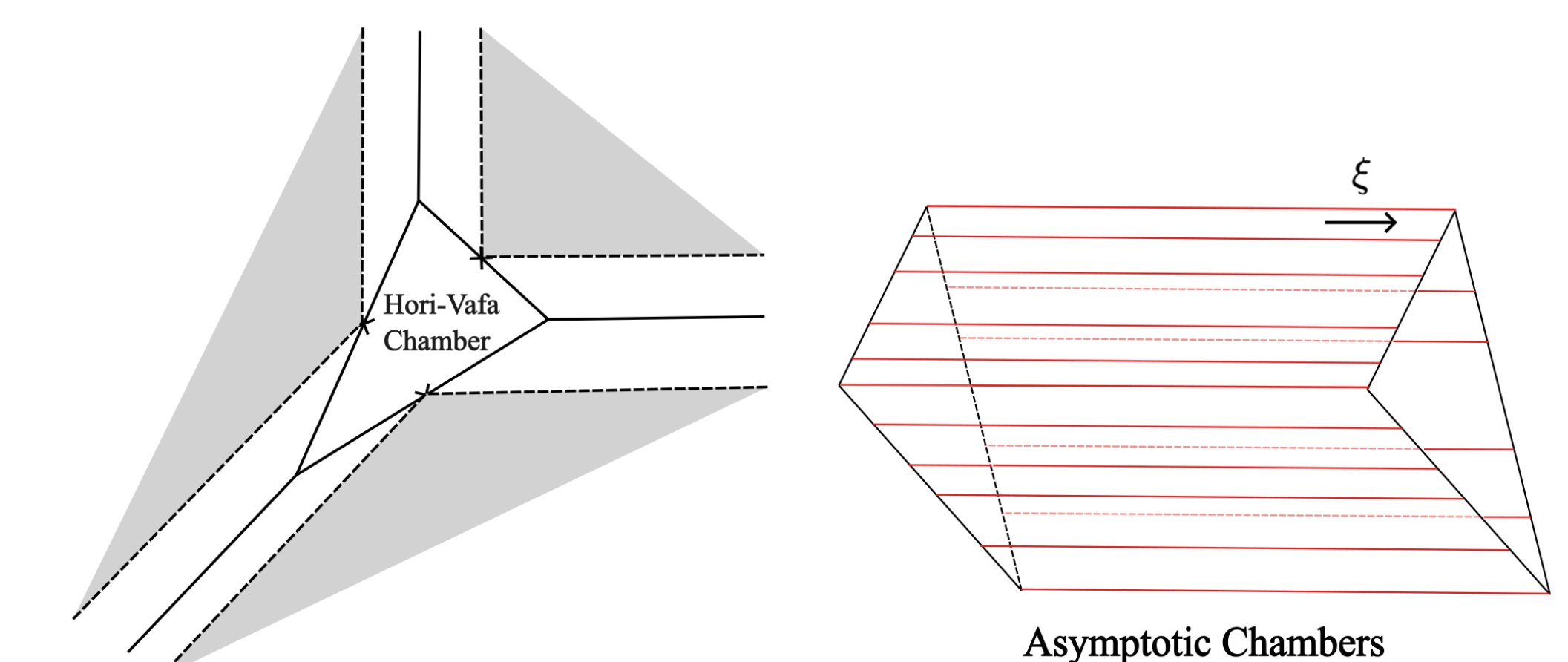
$$\mathcal{M}_{LG} := \mathcal{A}_{T_{HV}} / \mathcal{A}_{T_{HV}}(-1 + \widehat{W}_{HV})$$

Intrinsic Invariants from $(\mathcal{A}_{T_{HV}}, \mathcal{M}_{LG})$:

- Classification of DQ Algebroids $\mathcal{A}_{T_{HV}}$: Deligne-Fedosov Class $\mathfrak{D}(\mathcal{A}_{T_{HV}}) \in H^2(T_{HV}; \mathbb{C}_h^*)$
- Classification of DQ Pairs $(\mathcal{A}_{T_{HV}}, \mathcal{M}_{LG})$: Relative Deligne-Fedosov Class $\mathfrak{D}(\mathcal{A}_{T_{HV}}, \mathcal{M}_{LG}) \in H^2(T_{HV}, C; \mathbb{C}_h^*)$
- Intrinsic Quantum Period $QC(\mathcal{M}_{LG})$:
 $H^1(C; \mathbb{C}_h^*) \rightarrow H^2(T_{HV}, C; \mathbb{C}_h^*) \rightarrow H^2(T_{HV}; \mathbb{C}_h^*)$
 $QC(\mathcal{M}_{LG}) \mapsto \mathfrak{D}(\mathcal{A}_{T_{HV}}, \mathcal{M}_{LG}) \mapsto \mathfrak{D}(\mathcal{A}_{T_{HV}}) = 1$

Proper Landau-Ginzburg Models in Gross-Siebert Program [Carl-Pumperla-Siebert]

Completion of Hori-Vafa Landau-Ginzburg Model: Mirror Proper LG Model $(X, W) \supseteq (T_{HV}, W_{HV})$ of Toric Degeneration of $(\mathbb{P}^2, E)$ together with Its Deformation Quantization $(\mathcal{A}_X, \widehat{W})$ by [Bousseau].



Enumerative Meaning of Asymptotic Quantum Walls is Exactly the Desired A-Model Counting!